A seed is worth more than two layers

N=48 leak-free, ff=1024, 240000 x 100ep -- 50691430 / 50691432 / 50728837

The question

Finding 40 put L=10 (0.8007) BELOW L=8 (0.8660) on the shared n=1500 pool, paired p=5.4e-09. That test compares two CHECKPOINTS on the same instances; it says nothing about two TRAINING PROCEDURES, and no rung of this ladder had ever been run twice. H1 said the inversion was optimization noise. These two jobs retrain L=8 and L=10 with --seed 1 and change nothing else.

The answer: H1, and much more of it than predicted

At FIXED architecture the two seeds differ by L=8: 0.1627 (p=5.2e-42) L=10: 0.0947 (p=8.0e-17). The pre-registration called anything at or above 0.065 the alarm case, and said it was the outcome I did not expect. The measured spread is 0.1627 -- larger than the L=12 minus L=8 gap (0.1120) that finding 38's headline rests on, and 2.5x the inversion (0.0653) it was invoked to explain.

So the ladder cannot be read rung by rung -- and finding 40 is withdrawn

Every L* in this file (findings 11, 14, 22, 33, 38, 40) is a single seed per rung. A ladder whose rungs carry an unmeasured spread bigger than its steps cannot locate a knee, and the MLP-depth scan that was to be centred on L=10 cannot start until each rung is an average over seeds rather than a draw. The inversion itself did not survive: run within seed 1 the same comparison goes the OTHER way, L=10 - L=8 = +0.1920 (334-46, p=4.7e-55) against -0.0653 at seed 0. Finding 40 was a draw, not a property of depth 10.

The counterpoint: the SAMPLING error bar is now small

Job 50728837 re-evaluated the same six seed-0 checkpoints on a 4x larger depth pool (n_pool 6000, nested in the old 1500). L*(48, rho=0.60)=12's binding bin went n=70 -> 289 and its simultaneous 95% lower bound 0.8422 -> 0.9434, so it clears the gate at 95% for the first time. Sampling was never the big error bar here. The seed is.

Metric of record: sudoku_agg_eval, n=1500 at seed_start 5e10, disjoint from every training set, paired McNemar exact.
`seed` is not in INSTANCE_KEYS, so seed-1 cells score on exactly the same instances as seed-0 cells.

What the model actually sees

Every token is one categorical id into a 58-symbol vocabulary (N=48, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	`BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING`
	`BEGIN_ROUND SOLVED UNRESOLVED`
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	`COLOR_0 COLOR_1 COLOR_2 COLOR_3`
	the q colours (token name COLOR_n, prose 'colour')
10-57	`N0 N1 ... N47`
	one id per node, no structure in the id

GRAMMAR

`BEGIN_HINTS`	`(NODE COLOR)*`
`BEGIN_GRAPH`	`(NODE (NODE){degree})*`
`BEGIN_REASONING`	`(BEGIN_ROUND (NODE COLOR)+)*`

`SOLVED`

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image:
node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 393 TOKENS

HINTS	`BEGIN_HINTS N0 COLOR_1 N2 COLOR_2 N3 COLOR_3 N5 COLOR_0 N8 COLOR_2 N9 COLOR_2 ... (59 tokens)`
GRAPH	`BEGIN_GRAPH N0 N10 N22 N7 N9 N44 N1 N39 N27 N23 N40 N21 ... (289 tokens)`
REASONING	`BEGIN_REASONING BEGIN_ROUND N1 COLOR_1 N4 COLOR_3 N7 COLOR_0 N16 COLOR_2 N17 COLOR_3 N18 ... (45 tokens)`

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 5 rounds, 19 deductions,
38 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000023.

One real reasoning trace

29 hints of 48 nodes, then 5 rounds of simultaneous deduction. Hints: N0=COLOR_1 N2=COLOR_2 N3=COLOR_3 N5=COLOR_0 N8=COLOR_2 N9=COLOR_2 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND          N1 COLOR_1 N4 COLOR_3 N7 COLOR_0 N16 COLOR_2 ...
  10 deduced
BEGIN_ROUND          N6 COLOR_0 N24 COLOR_0 N27 COLOR_3 N31 COLOR_0 ...
  5 deduced
BEGIN_ROUND          N26 COLOR_2 N41 COLOR_1
  2 deduced
BEGIN_ROUND          N10 COLOR_3
  1 deduced
... 1 further rounds

SOLVED
```

WHY ONE DEDUCTION FIRES

node N1, neighbours N39 N27 N23 N40 N21

their colours known before this round:

```
   [0, 2, 3]  (3 = q-1 of the 4)

   (4 known neighbours, 3 DISTINCT colours -- some repeat)

   still unknown: N27
```

only colour left: 1

```
so the trace emits  N1 COLOR_1
```

The model is never shown this reasoning -- only the node/colour pair.

A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

N=48, degree 5, 4 colours, --shuffle-adj: the leak-free task. Every cell in this deck trains on exactly this encoding.

T=5 rounds, the 90th percentile of 40 draws from this config (range 1-9) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

The ladder, now with a second draw at two rungs

shared pool n=1500; Wilson 95%; val_nll from each cell's own history.json

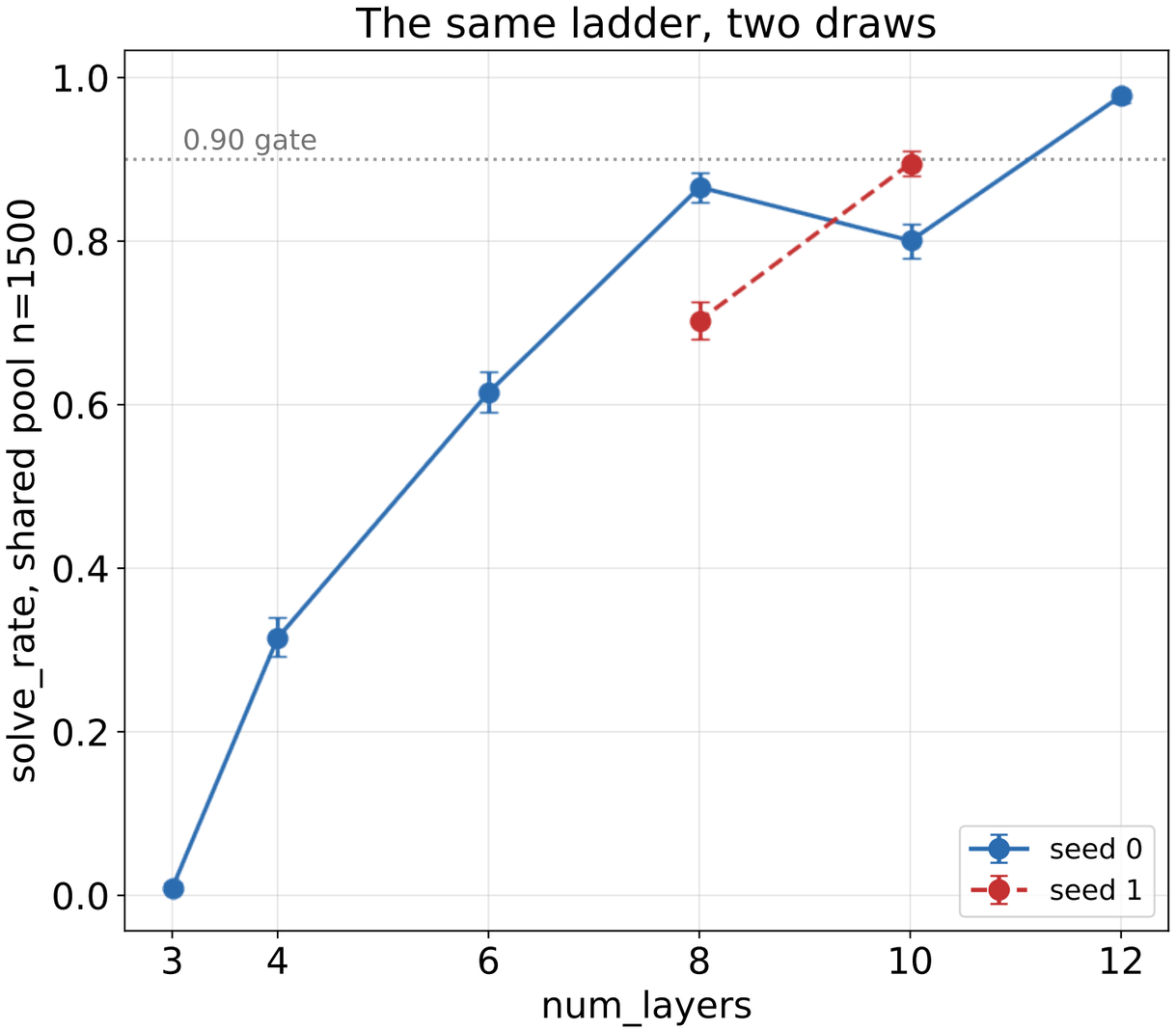
cell	solve	95% CI	val_nll	tok acc	trunc	out acc
L=12 s0	0.9780	[0.9693, 0.9843]	1.4993	0.4967	0.0007	0.9987
L=10 s0	0.8007	[0.7797, 0.8201]	1.5059	0.4942	0.0007	0.9953
L=10 s1	0.8953	[0.8788, 0.9098]	1.5013	0.4955	0.0000	1.0000
L=8 s0	0.8660	[0.8478, 0.8823]	1.5033	0.4947	0.0000	1.0000
L=8 s1	0.7033	[0.6797, 0.7259]	1.5084	0.4922	0.0007	0.9953
L=6 s0	0.6153	[0.5904, 0.6396]	1.5129	0.4921	0.0000	0.9960
L=4 s0	0.3153	[0.2923, 0.3393]	1.5422	0.4827	0.0000	1.0000
L=3 s0	0.0093	[0.0056, 0.0156]	1.5943	0.4608	0.0000	1.0000

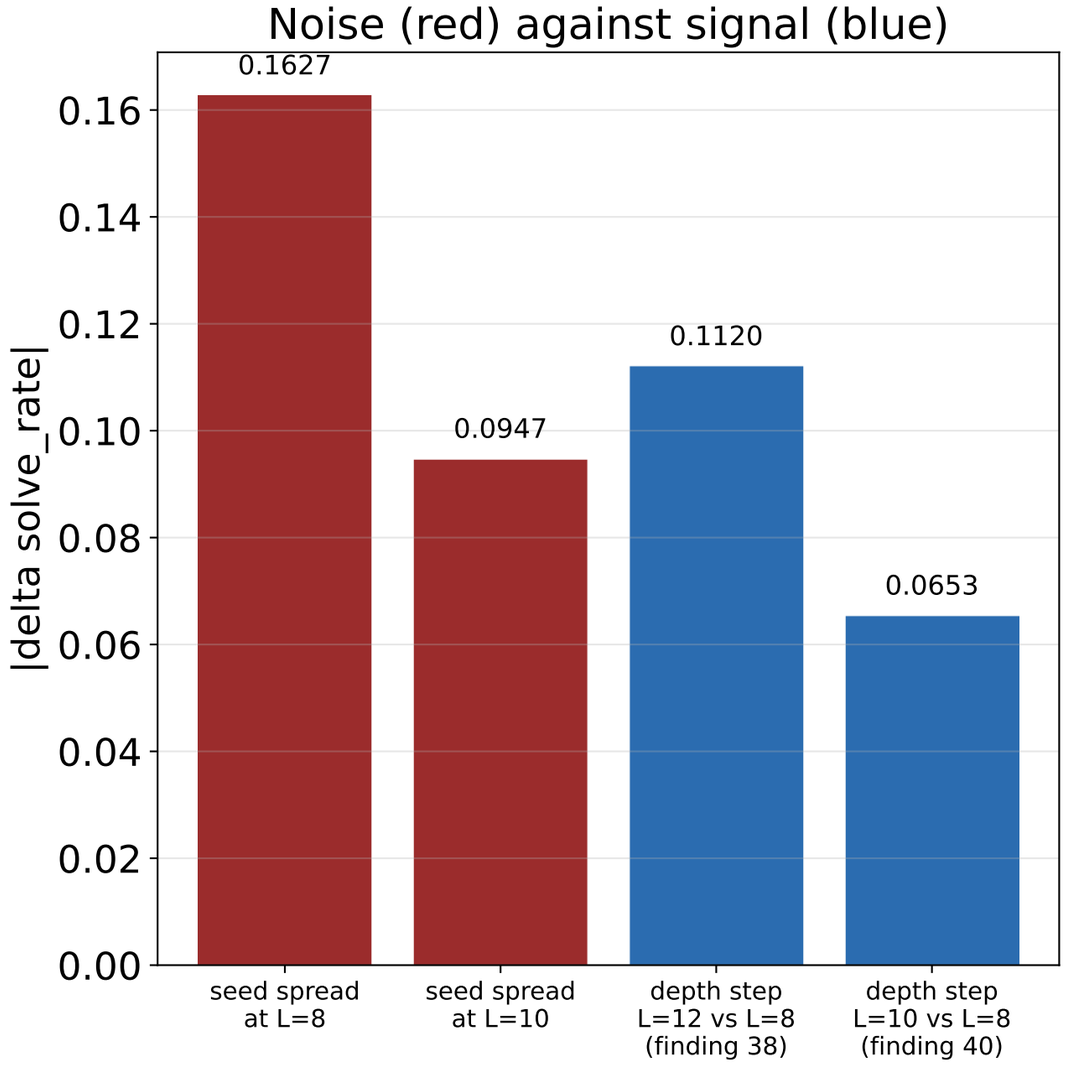
contrast	cells	kind	delta	disc	McNemar p
seed spread at L=8	8s1-8s0	noise	-0.1627	54-298	5.15e-42
seed spread at L=10	10s1-10s0	noise	+0.0947	220-78	7.97e-17
depth gap, finding 38	12s0-8s0	signal	+0.1120	182-14	1.90e-38
finding-40 inversion	10s0-8s0	signal	-0.0653	92-190	5.41e-09
depth, within seed 1	10s1-8s1	signal	+0.1920	334-46	4.68e-55

Every row is the same 1500 held-out instances, so each delta is a paired within-instance contrast, not two independent rates. 17 repeat evaluations of the seed-0 reference cells across four jobs agree instance for instance.

Reseeding one rung moves it further than changing its depth

shared n=1500 pool, Wilson 95%; the only difference between a red and a blue point at the same L is --seed

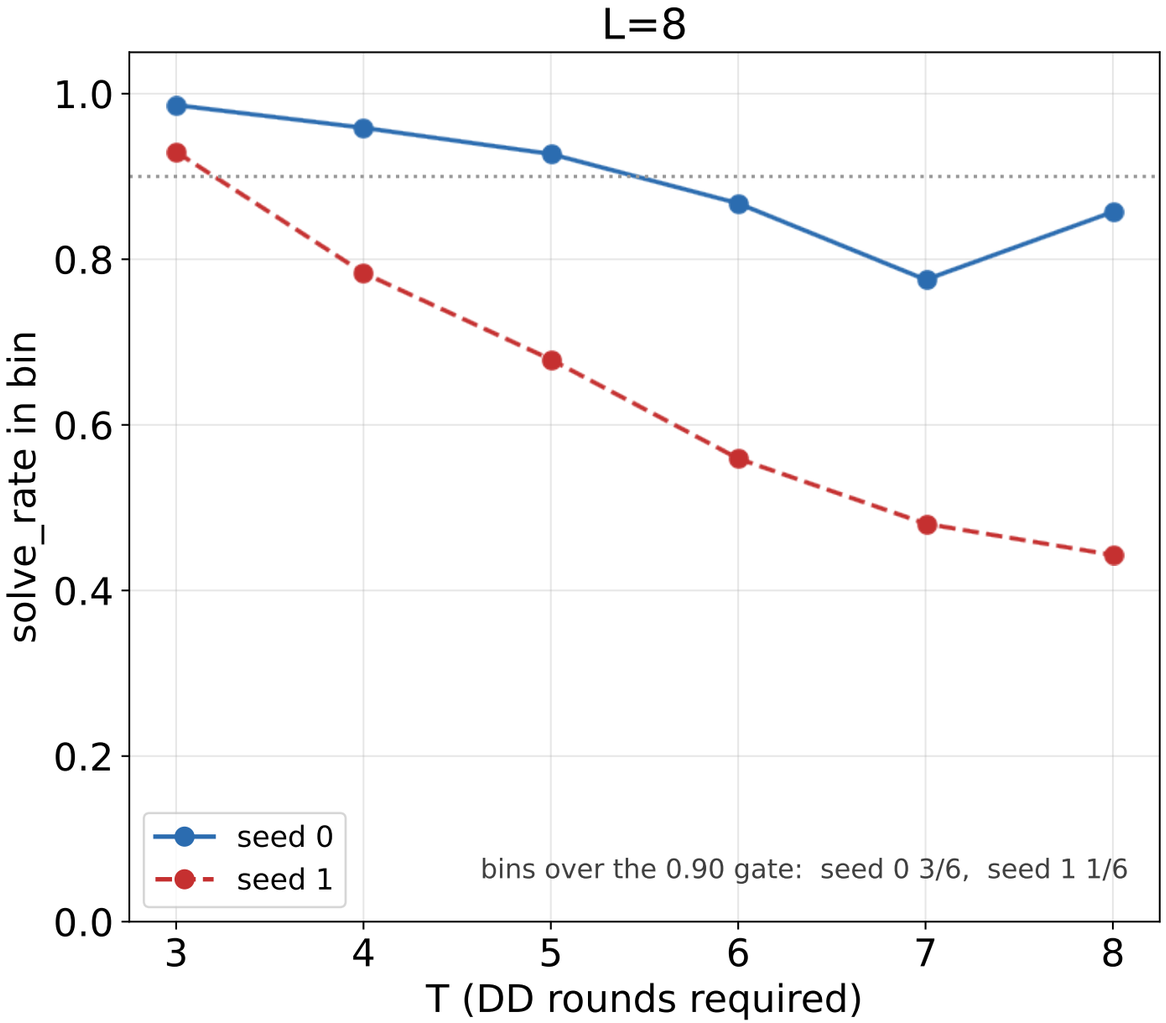


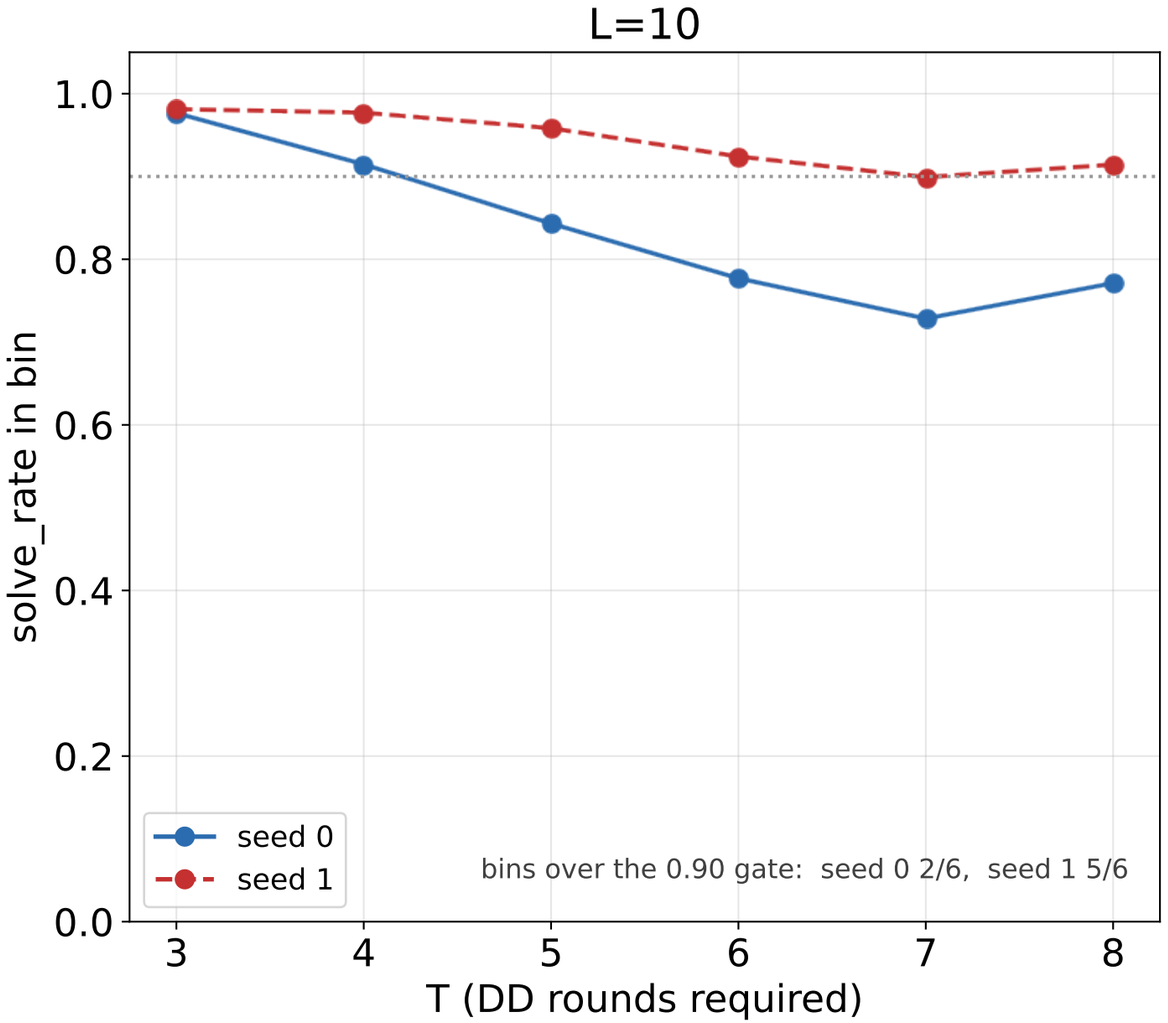


Red bars change NOTHING but the random seed; blue bars are the depth effects this ladder was built to measure. When the red bars are the taller ones, a single-seed rung is not a measurement of its architecture.

The seed acts where L* is read: the deep bins

rho=0.60, realized density pinned at 0.5836 in every bin (spread < 0.0005), so 'deeper' is not 'sparser'

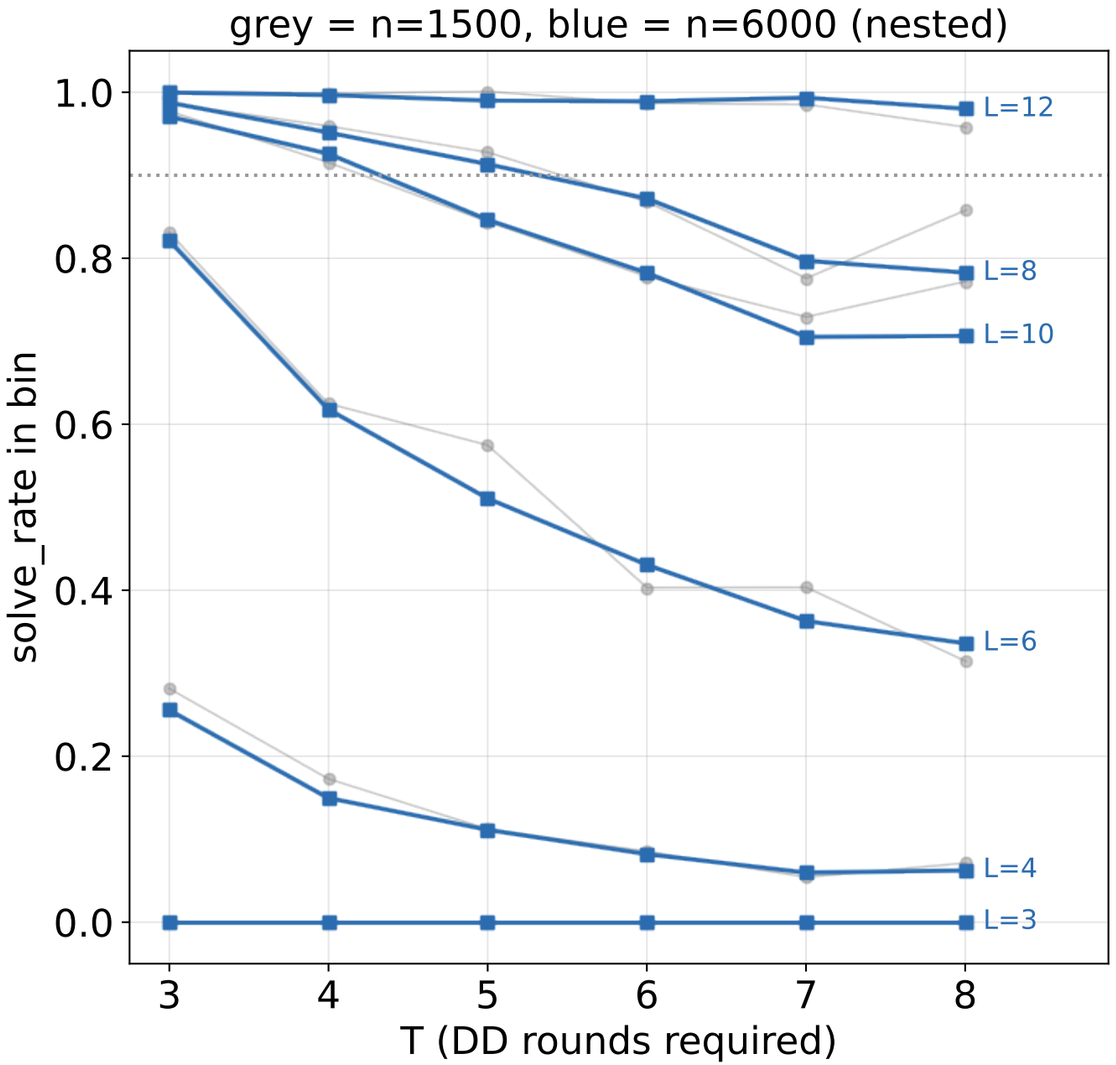




Note the two panels point opposite ways -- seed 1 is the worse draw at L=8 and the better one at L=10 -- but in both the seeds agree at T=3 and separate as T grows. The seed's effect is concentrated on exactly the axis L* is read off, which is why it can move L* by more than a layer can, in either direction.

The other error bar: sampling, and why it was not the problem

50728837 -- same six seed-0 checkpoints, depth pool 1500 -> 6000, no retraining




```
L*(48, leak-free, rho=0.60) = 12, binding bin T=8

                    n     rate   95% lo   gate at 95%
n_pool 1500        70   0.9571   0.8422   no
n_pool 6000       289   0.9792   0.9434   YES

Bonferroni-corrected Wilson over the six gated bins,
worst bin -- not the bin with the lowest point rate.
Nesting control, pre-registered: bin-size ratios
3.82-4.45 vs an expected 4.00, no bin's solve_rate
moved more than 2.54 SE. The pools nest exactly.

SAME AUDIT, EVERY L* IN STATUS (lstar_fragility.py,
CPU-only, reads CSVs already on disk):
  of the five L* values quoted flat, ONE held before
  tonight -- N=48 rho=0.75, L*=8 (lo 0.9455).
  N=96 rho=0.75 ff=1024 L*=4 clears 155/172 = 0.9012;
  at 154/172 it does not. ONE INSTANCE sets it, and
  finding 26's 'ff matters more at larger N' is 2.0x
  on points but 1.0x at 95%.
  Finding 31's L=7 misses by 0.0029 = 0.08 bin SE.

Why this hid: every other number here is a PAIRED
contrast on shared instances (finding 40: p=5.4e-09
on a 0.065 gap). L* is an ABSOLUTE threshold on ONE
cell, so it gets none of that power -- bin size is
all it has, and the deep bins were n=43-70.
```

Design rule from inverting the bound: at a simultaneous 95% over six bins, a cell whose true rate is 0.94 needs n=373, 0.96 needs n=161, 0.98 needs n=99. Cells at the knee sit at 0.93-0.96, so n=70 could never have cleared.

Pre-registration scorecard, and what to do about it

scored against scripts/slurm/sudoku2_n48_seed1_replicate.sbatch, written before either cell was submitted

(a) seed spread >= 0.040 :: CONFIRMED, in the alarm branch

max spread 0.1627 over L=8 (0.1627, 54-298, p=5.2e-42) and L=10 (0.0947, 220-78, p=8.0e-17). The header named >= 0.065 the case in which every single-seed L* in STATUS becomes uninterpretable without replicates, and said it was not the one I expected. (b) ORDERING: D_1 = pool(L=10,s1) - pool(L=8,s1) = +0.1920 (334-46, p=4.7e-55), against D_0 = -0.0653 at seed 0. Predicted D_1 > 0: CONFIRMED -- so finding 40's L=10 < L=8 inversion does not replicate, it changes sign. One seed of it was an artifact.

(c) MECHANISM -- val_nll concordance, sensitivity in [10, 60] :: (d) CONTROLS CONFIRMED

Within seed 1, val_nll(L=10) - val_nll(L=8) = -0.0071 and the pool delta is +0.1920; signs AGREE. Implied sensitivity 27.0 solve_rate per unit nll, against the pre-registered band [10, 60] and the seed-0 pairs' 25-28. The seed-0 pair gives 25.2, inside the band; the seeds differ by 0.0051 val_nll at L=8, predicting ~0.13 of solve rate against 0.1627 measured. Controls: realized density 0.5834-0.5839 vs 0.5836 +- 0.001; truncated_rate <= 0.0007 and outcome_accuracy >= 0.9953 everywhere, so these are deduction scores and not formatting; 17 repeat evals agree instance for instance.

What it costs the plan, and the cheapest fix

The MLP-depth scan has L* as its dependent variable, and L* now carries two error bars: sampling, measured and controlled tonight (+-0.03 at the binding bin), and seed, measured at 0.16 and not controlled. The scan cannot resolve a one-rung effect against the second. Fix: three seeds at L=8 and L=12 only, the cells that bracket L* at rho=0.60 -- SE ~0.09 per rung, 4 jobs at 7-11 h.

What is NOT in doubt, and the honest negative

The seed moves a rung, not the project: L=12 (0.9780) beats L=6 (0.6153) by 0.36 and L=3 (0.0093) by 0.97, and findings 27/28/29 (the DD verdict) are interventions on fixed checkpoints, untouched. Nothing improved tonight: the best N=48 leak-free cell is still L=12 seed 0 at 0.9780 [0.9693, 0.9843], unchanged since finding 38. The deliverable is an error bar, not a number.

Commit at submit a10a098, seed 1, N=48 leak-free ff=1024, 240000 x 100ep, --shuffle-adj; the only config difference from the seed-0 cells is --seed. Previous best on this eval: L=12 seed 0, 0.9780 (finding 38, job 50527633).